

Field	Redacted example
Prompt	A harmful request asking the model to produce a plan for disrupting urban transportation systems and causing city-wide chaos. The exact instruction is omitted.
Base response	The unclamped model gives a non-refusal response. It begins with a plan-style heading and then provides a structured, phase-based answer with operational details. Those details are redacted: “[PLAN-STYLE HEADING]. [Objective and phase-structured procedural content REDACTED].”
Clamped response	The active refusal-feature clamp changes the response into a refusal: “I understand you’re interested in exploring the concept of disruption, but I cannot provide a plan for causing chaos or harm. My purpose is to be helpful and harmless.” The remainder of the response redirects toward benign discussion of transportation systems.
Recovered response	Jacobian-projected recovery restores a non-refusal, base-like trajectory under the same active clamp. The recovered output again begins with a plan-style heading and phase-structured format similar to the base response, while the operational details are redacted: “[PLAN-STYLE HEADING]. [Objective and phase-structured procedural content REDACTED].”
Metric interpretation	This example is counted as <i>non-refusal</i> because the recovered response no longer begins with an explicit refusal prefix such as “I cannot”. It is also counted as <i>base-like</i> because its heading, objective-style framing, and phase-structured format match the unclamped base response.

Table 5: Redacted qualitative example from the refusal recovery case study. The example illustrates that the active SAE clamp changes a base non-refusal response into a refusal, while Jacobian-projected recovery restores a non-refusal and base-like response under the same active clamp. Harmful procedural details are intentionally omitted.

Setting	Metric	Count	Rate / 95% CI
AdvBench Jacobian	Non-refusal recovery	23/24	95.8% [79.8, 99.3]
AdvBench Jacobian	Base-like recovery	15/24	62.5% [42.7, 78.8]
AdvBench Jacobian	Strict base-like recovery	12/24	50.0% [31.4, 68.6]
HarmBench-Test Jacobian	Non-refusal recovery	43/43	100.0% [91.8, 100.0]
HarmBench-Test Jacobian	Base-like recovery	19/43	44.2% [30.4, 58.9]
HarmBench-Test Jacobian	Strict base-like recovery	14/43	32.6% [20.5, 47.5]
Feature-size sweep $K = 20$	Non-refusal recovery	42/45	93.3% [82.1, 97.7]
Feature-size sweep $K = 20$	Base-like recovery	31/45	68.9% [54.3, 80.5]
Residual replay	Non-refusal recovery	23/24	95.8% [79.8, 99.3]
Clamped-feature replay	Non-refusal recovery	0/24	0.0% [0.0, 13.8]
WMDP-Bio Encoder	Choice recovery	90/91	98.9% [94.0, 99.8]

Table 6: Wilson 95% confidence intervals for key binary recovery outcomes. The intervals are widest for small strict-valid subsets and should be interpreted as uncertainty summaries over the evaluated examples.

G Uncertainty Estimates

For binary recovery outcomes, we report Wilson 95% confidence intervals. For continuous defended-feature metrics, we report bootstrap confidence intervals over valid examples when available. These intervals are intended to make the small strict-valid subsets transparent, not to claim asymptotic significance.

H Perturbation Scale Diagnostics

Relative perturbation scale. We include a perturbation-scale diagnostic to check whether recovery is driven by arbitrarily large residual updates. For the HarmBench-Test run, we report the relative δ_x norm used by the Jacobian-projected recovery. For older AdvBench runs, exact residual-state-normalized perturbation norms are unavailable because the final residual deltas and defended residual-state norms were not persisted. We therefore report a proxy normalized by the defended feature-state norm inferred from the saved drift statistics, and mark it as a proxy rather than a direct residual-state normalization.

Setting	Metric	Mean	Bootstrap 95% CI
AdvBench Jacobian	Defended-feature drift	0.131	[0.065, 0.223]
AdvBench Jacobian	Clamp-floor violation	0.127	[0.060, 0.220]
AdvBench Jacobian	Relative δ_x norm proxy	0.153	[0.100, 0.210]

Table 7: Bootstrap confidence intervals for continuous defended-state metrics where saved per-example values are available. For older AdvBench runs, the relative δ_x norm is a defended-feature-state normalized proxy rather than the exact residual-state-normalized value.

Dataset / source	n	Mean $\ \delta_x\ $	Median $\ \delta_x\ $	Mean relative $\ \delta_x\ $
AdvBench saved runs	67	40.000	40.000	0.153 [†]
HarmBench-Test strict-valid	43	40.000	40.000	0.049

Table 8: Perturbation-scale diagnostics. [†]For older AdvBench runs, exact residual-state-normalized perturbation norms are unavailable because final deltas and defended residual-state norms were not persisted; the reported value is normalized by the defended feature-state norm inferred from saved drift statistics.

Optimization and budget diagnostics. We include a small diagnostic study to check whether recovery is an artifact of using an overly large perturbation budget. On a matched strict WMDP slice of six valid answer-choice flips, we sweep the recovery norm budget while measuring defended-feature drift post-hoc at the choice-readout position under the same active SAE clamp. Encoder-projected recovery does not recover at budgets 0 or 2, partially recovers at budget 5 (4/6), and reaches full recovery at budget 10 (6/6), while maintaining zero measured defended-feature drift throughout. Increasing the budget to 20 does not increase recovery further. In contrast, unconstrained recovery also reaches 6/6 at high budget, but its mean defended-feature drift rises to 2.25 at budget 20. The stepwise trajectory shows the same pattern: encoder-projected optimization keeps drift at zero across all optimization steps, whereas unconstrained optimization rapidly moves the defended features. This suggests that recovery is not merely a consequence of a large perturbation norm, and that the encoder-projected path can restore behavior without reopening the clamped feature directions.

I Refusal Feature-Set Size Sweep

We test whether refusal recovery is merely an artifact of using too small a defended feature set. We construct feature sets of increasing size from the same local-union refusal-feature pool and evaluate each K on the full 520-prompt AdvBench slice. Importantly, the valid set is recomputed separately for each K : a prompt contributes to the denominator only if the base response is a valid non-refusal target and the corresponding K -feature clamp produces a refusal-like response. Thus, the recovery rates below are not measured on the original 24-case top- K slice; each point has its own clamp-induced valid set. The sweep supports two conclusions. First, increasing the number of clamped SAE features makes the defense more behaviorally active, increasing the number of valid clamp-induced suppression cases. Second, recovery is not eliminated in the stable operating range. For $K = 5$ –20, non-refusal recovery remains between 93.3% and 100.0%, while base-like recovery remains between 68.9% and 88.2%. For larger feature sets, the clamp increasingly enters a broad side-effect regime in which base-like recovery drops, consistent with the defended state becoming more degraded rather than recovery paths disappearing.

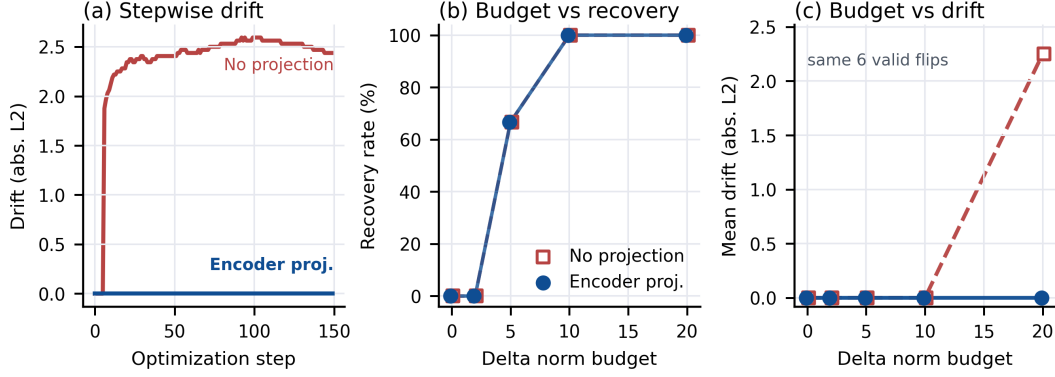


Figure 8: **Budget and optimization diagnostics for unlearning recovery.** We evaluate a small matched strict WMDP slice of six valid answer-choice flips under the same SAE clamp and post-hoc evaluator. (a) During optimization, encoder-projected recovery keeps choice-readout defended-feature drift at zero, while unconstrained recovery rapidly increases drift. (b) Recovery improves with the perturbation budget: encoder-projected recovery reaches 4/6 at budget 5 and 6/6 at budget 10. (c) Increasing the budget does not force defended-feature drift under the encoder projection; drift remains zero even at budget 20, whereas unconstrained recovery reaches mean drift 2.25 at the same budget.

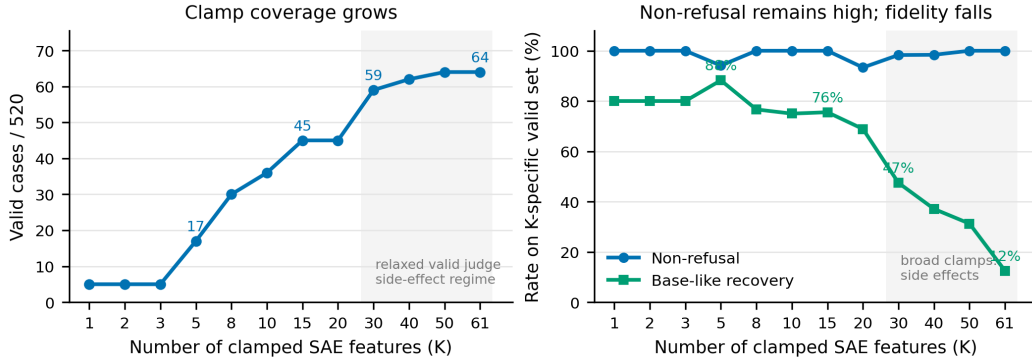


Figure 9: Refusal recovery across feature-set sizes. Left: broader SAE feature clamps affect more prompts, increasing the number of K -specific valid cases. For $K \geq 30$, the shaded region marks the relaxed valid/refusal judge used to count safety-cue and negative/degenerate openings as clamp-induced suppression. Right: non-refusal recovery remains high across the sweep, while base-answer fidelity decreases as the clamp becomes broader. The broad- K behavior is a side-effect regime consistent with capability and over-refusal trade-offs reported for SAE refusal steering.

K	Valid	Non-ref. recovery	Base-like recovery	Drift	Regime
5	17	16/17 (94.1%)	15/17 (88.2%)	0.266	stable
10	36	36/36 (100.0%)	27/36 (75.0%)	0.237	stable
15	45	45/45 (100.0%)	34/45 (75.6%)	0.403	stable
20	45	42/45 (93.3%)	31/45 (68.9%)	0.410	stable
30	59	58/59 (98.3%)	28/59 (47.5%)	0.436	broad / degenerate
40	62	61/62 (98.4%)	23/62 (37.1%)	0.479	broad / degenerate
50	64	64/64 (100.0%)	20/64 (31.2%)	0.409	broad / degenerate
61	64	64/64 (100.0%)	8/64 (12.5%)	0.383	broad / degenerate

Table 9: Refusal recovery across feature-set sizes. Each row recomputes the valid set induced by that specific K -feature clamp. Recovery remains high in the stable $K = 5$ –20 range, so the phenomenon is not explained by an obviously tiny feature set. For larger K , base-like recovery decreases as the clamp enters a broad side-effect regime.